

MATH 3115 MSC & SED COMPLEX ANALYSIS

Lecture 1: Introduction to complex numbers

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Lesson Outcome

By the end of this lesson you should be able to:

- Describe numbers in a real number line.
- Define the complex numbers.
- Add complex numbers
- Subtract Complex numbers

INTRODUCTION

- Complex numbers are a powerful mathematical tool that extends our understanding of the physical world.
- They enable us to solve problems that would be difficult or impossible to address with only real numbers.
- They have broad applications in both theoretical and practical domains.
- In this lecture we discuss about the number system

NUMBER SYSTEM

Definition 1 (Natural numbers)

These are the numbers that were first used in counting. They are also called counting numbers or positive integer .

- They are denoted by $\mathbb{N} = \{1, 2, 3, \dots\}$
- If $a, b \in \mathbb{N}$, then $a + b \in \mathbb{N}$, and $ab \in \mathbb{N}$ while $a - b \notin \mathbb{N}$ and $\frac{a}{b} \notin \mathbb{N}$.
- $3 - 2 = 1 \in \mathbb{N}$ while $2 - 3 = -1 \notin \mathbb{N}$.
- Also $\frac{6}{2} = 3 \in \mathbb{N}$ while $\frac{2}{6} = \frac{1}{3} \notin \mathbb{N}$.

NUMBER SYSTEM CONT'D

Definition 2 (Whole numbers)

These are composed of natural numbers including zero.

- They are denoted by $\mathbb{W} = \{0, 1, 2, \dots\}$.
- The system permits subtraction by the same number.
- Closed under addition, multiplication and subtraction by the same numbers

NUMBER SYSTEM CONT'D

Definition 3 (Integers)

These are whole numbers and negative of the natural numbers.

- They are denoted denoted by $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, \dots\}$.
- They permit solutions of equation such as $x + b = a$, where $a, b \in \mathbb{N}$.
- And we can write $x = a - b \in \mathbb{Z}$.
- Closed under addition, subtraction and multiplication.

NUMBER SYSTEM CONT'D

Definition 4 (Rational numbers)

These are numbers which can be expressed in form of a fraction.

- They are denoted by $\mathbb{Q} = \{\frac{a}{b} \mid a, b \in \mathbb{Z}\}$.
- They permits solutions of the equation $bx = a$, $b \neq 0$. That is $x = \frac{a}{b}$.
- Closed under addition, subtraction, multiplication and division excluding division by zero.

NUMBER SYSTEM CONT'D

Definition 5 (Irrational numbers)

These are numbers that cannot be expressed in the form of a fraction,

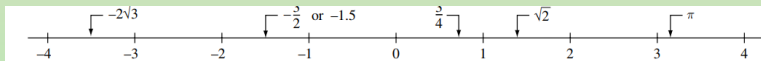
- They are denoted by $\mathbb{Q}'$.
- For example, $\sqrt{2}$, π , and e are irrational numbers.

NUMBER SYSTEM CONT'D

Definition 6 (Real numbers)

These form the set of rational numbers and irrational numbers and denoted by $\mathbb{R}$.

- They are represented graphically by points on the real axis.
- Where the point corresponding to zero is called the Origin.



- For two numbers on the real line we can write $a < b$, $a = b$ and $a > b$

Complex numbers

Definition 7

A complex number is an expression of the form $z = a + bi$ where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$ or $i^2 = -1$.

- The number a is the real part of the complex number while b is the imaginary part.
- If $z = a + bi$, then $a = \operatorname{Re}(z)$ and $b = \operatorname{Im}(z)$.

Definition 8 (Equality of Complex Numbers)

Two complex numbers $z = a + bi$ and $w = c + di$ are equal if and only if $a = c$ and $b = d$.

Complex numbers

Definition 9

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Algebra of Complex numbers

Theorem 11

Suppose that z, w , and v belong to the set $\mathbb{C}$ of complex numbers, then

- (a) $z + w$ and zw belong to $\mathbb{C}$ closure property
- (b) $z + w = w + z$ Commutative property of addition
- (c) $(z + w) + v = z + (w + v)$ associative property of addition
- (d) $zw = wz$ Commutative property of multiplication
- (e) $(zw)v = z(wv)$ Associative property of multiplication
- (f) $z(w + v) = zw + zv$ Distributive property over addition
- (g) $z + 0 = 0 + z = z$ identity property of addition
- (h) $1 \cdot z = z \cdot 1 = z$ identity property of multiplication
- (i) For every complex number z , there is a unique number $-z \in \mathbb{C}$ such that $z + (-z) = 0$. z is called the additive inverse of multiplication denoted by $-z$
- (j) For any $z \neq 0$, there is a unique number $w \in \mathbb{C}$ such that $zw = wz = 1$. w is called multiplicative inverse of multiplication denoted by $\frac{1}{w}$ or w^{-1}

Algebra of Complex numbers CONT'D

proof of b.

Let $z = a + bi$ and $w = c + di$ then

$$\begin{aligned} z + w &= (a + bi) + (c + di) \\ &= a + bi + c + di \\ &= (a + c) + (bi + di) \text{ adding real and imaginary parts} \\ &= (c + a) + (di + bi) \text{ real numbers commute} \\ &= c + a + (d + b)i \text{ writing as a complex number} \\ &= c + di + a + bi \\ &= w + z \end{aligned}$$



Algebra of Complex numbers CONT'D

proof of G.

$$\begin{aligned} z &= a + bi, \text{ then} \\ z + 0 &= (a + bi) + (0 + 0i) \\ &= a + bi + 0 + 0i \\ &= a + 0 + bi + 0i && \text{adding real and imaginary part} \\ &= 0 + a + 0i + bi \\ &= 0 + 0i + a + bi && \text{writing as two complex numbers} \\ &= 0 + z \\ &= z. \end{aligned}$$



Algebra of Complex numbers CONT'D

Example 12

Let $z = 2 + 3i$ $w = 2 + 7i$ $v = 1 + i$, Find $z + w$.

Solution 13

$$\begin{aligned} z + w &= (2 + 3i) + (2 + 7i) \\ &= (2 + 3i) + (2 + 7i) \\ &= 2 + 3i + 2 + 7i \quad \text{add real and imaginary} \\ &= (2 + 2) + (3i + 7i) \\ &= 4 + 10i \end{aligned}$$

Algebra of Complex numbers CONT'D

Example 14

Given that $z = 3 - i$, $w = 5 + 2i$ and $v = 2 + 2i$, compute the following:

(a) $4z - 2v$

(b) zw

Solution 15

$$\begin{aligned}4z - 2v &= 4(3 - i) - 2(2 + 2i) \\&= 12 - 4i - 4 - 4i \\&= 12 - 4 - 4i - 4i \\&= 8 - 8i\end{aligned}$$

Algebra of Complex numbers CONT'D

Solution 16

$$\begin{aligned}zw &= (3 - i)(5 + 2i) \\&= 3 \cdot 5 + 3 \cdot 2i - i \cdot 5 - (i \cdot 2i) \\&= 15 + 6i - 5i - 2i^2 \\&= 15 + 6i - 5i - 2(-1), i^2 = -1 \\&= 15 + 6i - 5i + 2 \\&= 15 + 2 + 6i - 5i \\&= 17 + i\end{aligned}$$

Algebra of Complex numbers CONT'D

Example 17

Perform the required calculation and express the answer in the form of $a + bi$.

(a) $(3 - 2i)(3i + 5)$

Solution 18

$$\begin{aligned}(3 - 2i)(3i + 5) &= 3 \cdot 3i + 3 \cdot 5 - 2i \cdot 3i - 2i \cdot 5 \\&= 9i + 15 - 6i^2 - 10i \\&= 9i + 15 - 6(-1) - 10i \quad \text{but } i^2 = -1 \\&= 9i + 15 + 6 - 10i \\&= 15 + 6 + 9i - 10i \quad \text{like terms} \\&= 21 - i \quad \text{in the form of } a + bi\end{aligned}$$

Graph representation of complex numbers

- We display complex numbers on a Complex plane, or an Argand diagram or the z plane.
- The x axis is called the real axis because that is where real part of $\mathbb{C}$ is placed.
- The y axis is called the imaginary axis because imaginary numbers are placed on the y axis.

Graph of Complex numbers CONT'D

Example 19

Locate the following complex numbers on the complex plane

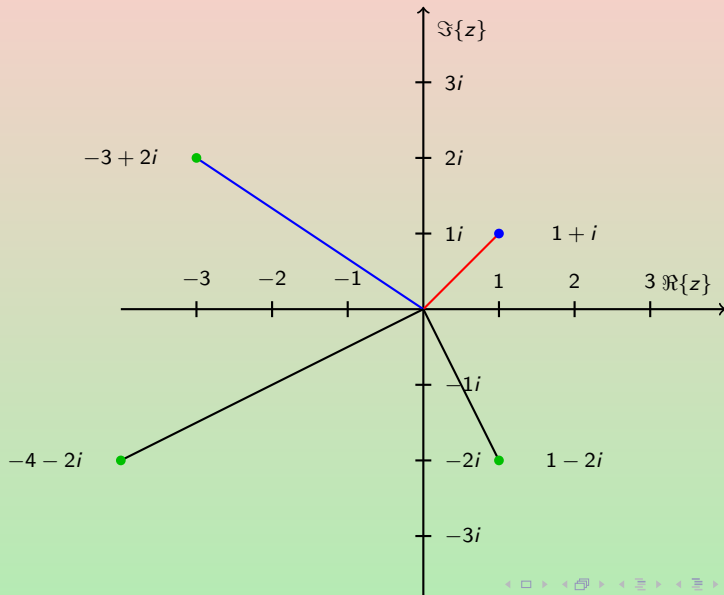
(a) $1 + i$

(b) $1 - 2i$

(c) $-3 + 2i$

(d) $-4 - 2i$

Argand diagram Example cont'd



Addition of complex numbers

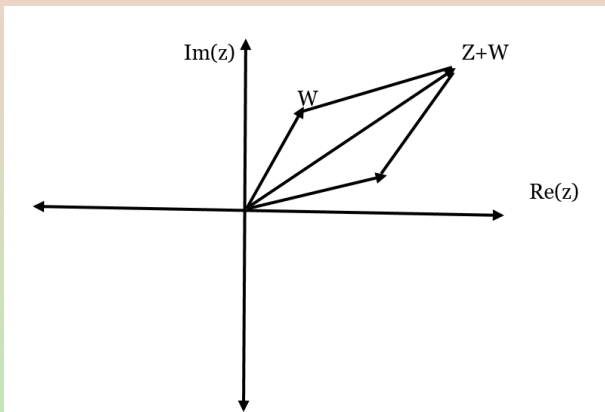
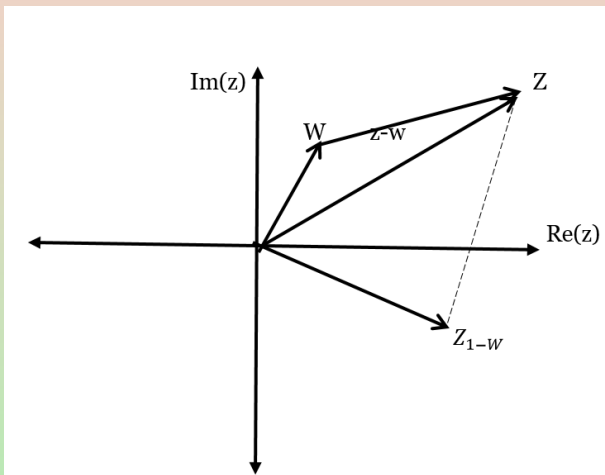


Figure: The sum $Z+W$

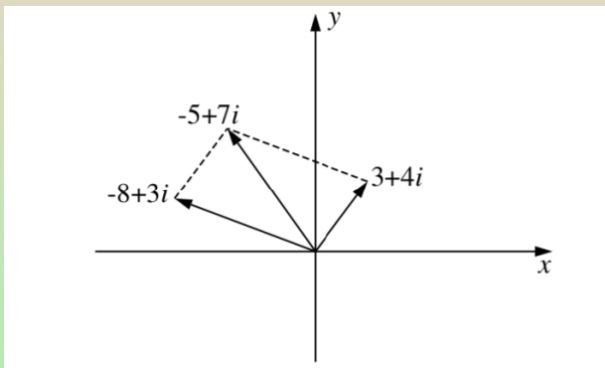
Addition of complex numbers



Addition of complex numbers

Example 20

Let $w = 3 + 4i$ and $z = -8 + 3i$. Use the vector form of complex numbers to represent geometrically $w + z$.



THE END